

Predicting the Unpredictable: AI Techniques for Supply Chain Risk Mitigation

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Abstract. The abstract should briefly summarise the contents of the paper in 150–250 words.

Keywords: Artificial Intelligence · Supply Chain · Risk Mitigation · Machine Learning

1 Introduction

This section introduces the topic of the paper, which is the application of artificial intelligence (AI) techniques for supply chain risk mitigation.

2 Theoretical Background

3 Methodology

This section describes the systematic literature review (SLR) protocol adopted to identify, screen, and synthesise the body of evidence concerning the use of Artificial Intelligence (AI) for risk mitigation in Supply Chain Management (SCM). The procedure was designed to ensure transparency, reproducibility, and methodological rigour throughout every stage of the review.

3.1 Review Protocol

The review followed the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses* (PRISMA) 2020 guidelines, which provide a structured framework for reporting the identification, screening, and inclusion of studies in systematic reviews.

The overarching goal of the protocol was to answer, in a systematic and reproducible manner, the following research questions (RQs):

- **RQ1:** How can Artificial Intelligence be applied to mitigate risks in supply chains?
- **RQ2:** Which types of supply chain risks are addressed through AI-based approaches?
- **RQ3:** Which Artificial Intelligence techniques are employed for supply chain risk mitigation?

Each phase of the review—identification, screening, eligibility, and inclusion—was documented in accordance with the PRISMA 2020 flow diagram, allowing the entire selection process to be audited and replicated.

3.2 Databases Searched

Two scientific databases were selected as primary sources for the literature search: **IEEE Xplore** and **Web of Science**.

The coverage period was restricted to publications released between **2019 and 2026**. This window was chosen to capture the most recent wave of AI adoption in SCM, including the period following major disruptions such as the COVID-19 pandemic and subsequent geopolitical shocks, which have substantially reshaped the academic discourse on supply chain resilience and risk.

3.3 Search Queries

The search string was constructed along three conceptual dimensions: (i) the application *domain* (Supply Chain), (ii) the *technology* of interest (Artificial Intelligence and its synonyms), and (iii) the *type of analysis* pursued (Risk Mitigation). Boolean operators were used to combine the dimensions, while quotation marks ensured exact-phrase matching. The final query is presented below:

"Supply Chain" AND ("Artificial Intelligence" OR "AI") AND "Risk Mitigation"

The query was applied to the title, abstract, and keyword fields of both databases. Table 1 summarises the number of records retrieved per database.

Table 1. Number of records retrieved per database.

Database	Records Retrieved
IEEE Xplore	142
Web of Science	50
Total	192

The rationale behind combining these three dimensions was to balance *recall*" (broad enough to capture different AI families, e.g., Machine Learning, Deep Learning, NLP) with *precision* (narrow enough to filter out studies that, while mentioning AI or SCM, do not address the risk dimension explicitly).

3.4 Screening Criteria

A first set of thematic filters was applied during the screening phase, that is, while reading the title, keywords, and abstract of each record. To advance beyond this stage, an article had to simultaneously satisfy the following three requirements:

1. **Exclusive Supply Chain context.** The study had to demonstrate a practical application within logistics, procurement, inventory management, production planning, or transport networks. Articles addressing supporting technologies such as Blockchain or IoT from a purely technical perspective (e.g., network architecture or communication protocols), without applying them to logistics operations, were rejected.
2. **Explicit presence of Artificial Intelligence.** Studies were required to employ advanced AI techniques, such as Machine Learning (ML), Neural Networks, Deep Learning (DL), Natural Language Processing (NLP), or genetic algorithms. Works limited to basic digitalisation, traditional automation, ERP systems, or the mere deployment of IoT/Blockchain sensors for data collection—without AI-based predictive or prescriptive processing—were excluded.
3. **Central focus on risk prevention and mitigation.** The AI component had to be deployed in order to anticipate disruptions, manage market uncertainty (supply or demand), build resilience, foresee supplier failures, or minimise the impact of adverse events. Applications targeting strictly operational or commercial goals (e.g., optimising daily delivery routes or sales forecasting for marketing purposes), without a clear risk or resilience dimension, were excluded.

3.5 Eligibility Criteria

Articles that passed the screening phase were then subjected to a more demanding set of inclusion and exclusion criteria, applied during the full-text reading. The objective at this stage was no longer to identify articles broadly related to AI and Supply Chain, but to retain only studies offering a *direct, substantive, and methodologically explicit* contribution to the research questions formulated in Section 3.1.

Inclusion Criteria. An article was retained only if it satisfied *all* of the following requirements:

- **IC1 — Centrality of AI:** Artificial Intelligence had to be a central element of the article, rather than a peripheral or merely cited concept.
- **IC2 — Risk-oriented purpose:** The main focus of the study had to be the mitigation, prediction, management, or assessment of risk in a supply chain context.
- **IC3 — Methodological clarity:** The article had to describe explicitly (i) the AI techniques employed, (ii) the type(s) of risk addressed, and (iii) a concrete application, model, or methodology.
- **IC4 — Tangible contribution:** The work had to present results, a framework, a model, an empirical study, or an applicable methodology.
- **IC5 — Alignment with the RQs:** The article had to provide an explicit contribution to at least one of the three research questions (RQ1, RQ2, or RQ3).

- **IC6 — Generalisability:** The context studied had to be sufficiently transferable to business and commercial supply chain settings.

Exclusion Criteria. Conversely, an article was excluded whenever *any* of the following situations occurred:

- **EC1 — Superficial use of AI:** AI mentioned without an underlying implementation or concrete methodology.
- **EC2 — Operational focus without risk:** Primary emphasis on optimisation, efficiency, logistics, or operational performance with no clear link to risk mitigation.
- **EC3 — AI as an accessory technology:** The article centres on digital transformation, Industry 4.0, IoT, Blockchain, or Big Data as its main theme, with AI playing only an ancillary role.
- **EC4 — Unclear risk typology:** Failure to clearly identify the type(s) of supply chain risk being addressed.
- **EC5 — Absence of a specific AI technique:** No specific AI model, algorithm, or method is described.
- **EC6 — Excessive abstraction:** Overly conceptual or generic articles, lacking applicable results.
- **EC7 — Out-of-scope risk:** The risk discussed is not genuinely a supply chain risk.
- **EC8 — Limited transferability:** Studies confined to highly idiosyncratic contexts, whose findings cannot reasonably be generalised to broader business settings.

3.6 Screening Process

The selection of studies was organised in three sequential phases, in line with the PRISMA 2020 flow.

Phase 1 — Identification and removal of duplicates. The 192 records exported from IEEE Xplore and Web of Science were consolidated into a single reference manager. Three duplicate entries were identified and removed, resulting in 189 unique records advancing to the next stage.

Phase 2 — Title, keywords, and abstract screening. Each remaining record was evaluated based on its title, keywords, and abstract, against the three thematic filters defined in Section 3.4. All three filters had to be simultaneously satisfied for an article to advance to the next phase. To ensure transparency and traceability, this stage was fully documented in a dedicated spreadsheet, in which each record was registered together with the following fields: *Title*, *Decision*, *Risk Analysed*, *AI Technique*, and *Justification*. This structure made it possible to revisit individual decisions, audit the rationale behind each exclusion, and verify the consistency of judgements across the dataset. At the end of this phase, 128 records were removed (23 from IEEE Xplore and 105 from Web of Science), leaving 61 studies eligible for full-text assessment.

Phase 3 — Full-text reading and eligibility assessment. The 61 surviving articles were retrieved in full and subjected to a stricter evaluation against the inclusion and exclusion criteria defined in Section 3.5. This phase was also documented in a structured spreadsheet, where each article was characterised through a richer set of descriptors aimed at supporting both the eligibility decision and the subsequent qualitative synthesis. For every record the following fields were registered: *Title, Authors, Year, Journal/Conference, Country of the Authors, Inclusion Criteria Satisfied, Exclusion Criteria Triggered, Decision, Justification of the Decision, AI Techniques, Types of Risk Addressed, Mitigation Approach, Application Sector, Type of Study, Answer to RQ1, Answer to RQ2, Answer to RQ3, Problem Addressed, Main Contribution, Limitations, and Future Work.*

This extended template fulfilled a dual purpose. On the one hand, it operationalised the eligibility assessment by forcing an explicit mapping between each article and the inclusion and exclusion criteria, as well as the perceived strength of relevance (high, medium, or low). On the other hand, it already constituted the basis for the data extraction stage, since the AI techniques, risk types, mitigation approaches, and contributions per research question would later feed directly into the qualitative synthesis. A conservative posture was deliberately adopted: in cases of doubt, or whenever the methodological description of either the AI component or the risk dimension was insufficient, the article was excluded. Studies classified as having only *low* relevance were likewise discarded, even when formally meeting the basic inclusion criteria.

Of the 61 articles assessed in this phase, 2 could not be retrieved in full text and 40 were excluded for failing to meet at least one of the criteria above, resulting in a final corpus of **19 studies** included in the qualitative synthesis.

Reviewer procedure. The screening and eligibility assessment were conducted as a team effort, with the set of articles distributed among the authors and evaluated individually against the criteria defined above. Each article was therefore reviewed by a single author, but the criteria, the spreadsheet templates, and the decision rubric were collectively defined and applied uniformly across the team. Borderline cases and decisions flagged as uncertain were brought to joint discussion and re-examined collectively, in order to ensure consistency of judgement across reviewers and to mitigate individual bias.

Figure 1 summarises the full selection process, from the 192 records initially identified to the 19 studies retained for analysis.

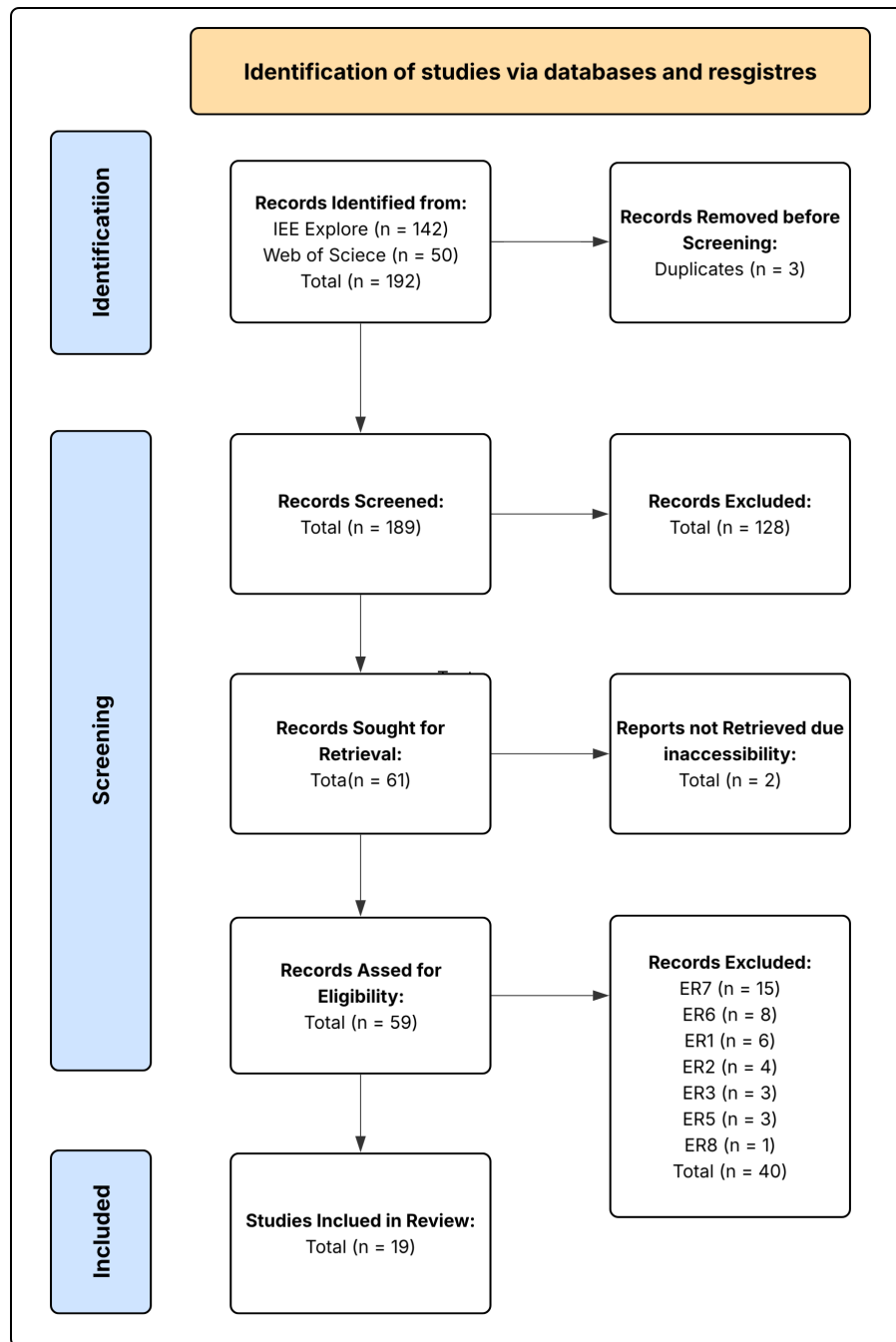


Fig. 1. PRISMA 2020 flow diagram of the study selection process.

4 Selection Results

Before addressing the research questions, a descriptive characterization of the 19 selected articles is presented. This analysis contextualizes the state of the field by examining the temporal evolution of publications, geographic concentration, methodological diversity, and sectoral coverage of the included studies.

4.1 Temporal Distribution

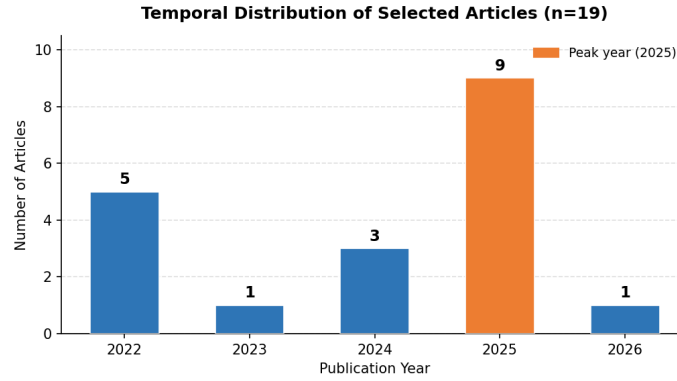


Fig. 2. Temporal distribution of selected articles (n=19).

The temporal distribution of the 19 selected articles (Fig. 2) reveals a clear acceleration in research activity at the intersection of Artificial Intelligence and Supply Chain Risk Management (SCRM) over recent years. The earliest cluster (2022) accounts for five publications, coinciding with a period of heightened academic interest driven by the operational disruptions caused by the COVID-19 pandemic and subsequent supply chain instability. A relative consolidation is observed in 2023 (n=1) and 2024 (n=3), suggesting a transitional period as the field matured and pivoted towards more technically sophisticated approaches.

The most notable finding is the sharp surge in 2025, which alone concentrates nine of the nineteen articles (47.4%), representing the single most productive year in the corpus. This trajectory strongly indicates that the application of AI to SCRM is an emergent and rapidly evolving domain, where foundational conceptual work (2022) is giving way to empirical validation and system deployment (2025–2026). The presence of one article from 2026 further corroborates that the research frontier is active and ongoing, underscoring the timeliness and relevance of this systematic review.

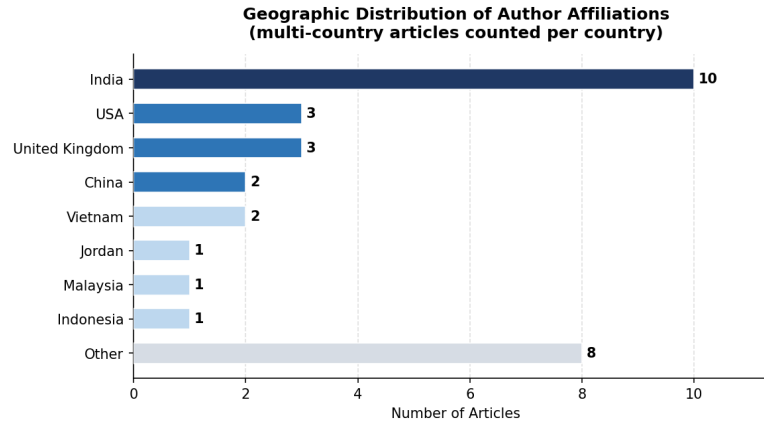


Fig. 3. Geographic distribution of author affiliations (multi-country articles counted per country, $n=19$).

4.2 Geographic Distribution

The geographic analysis (Fig. 3) reveals a pronounced concentration of research output originating from Asia and Anglo-Saxon academic contexts. India emerges as the dominant contributor, with affiliations appearing in ten of the nineteen articles, a figure that reflects both the country’s substantial investment in AI research capacity and the strategic importance of supply chain resilience in its manufacturing and logistics sectors. The United States ($n=3$) and United Kingdom ($n=3$) follow, consistent with their historically strong publication output in operations management and computer science. China ($n=2$) and Vietnam ($n=2$) are also represented, both nations with significant stakes in global manufacturing networks.

A critical observation is the near-complete absence of European and South American authorship. With the exception of one article from Germany and one from Hungary (both as co-authors in multi-country collaborations), the European research community appears largely underrepresented in this specific intersection of AI and SCRM. This constitutes a meaningful geographic gap, as European supply chains—heavily regulated and deeply integrated through European Union frameworks—may face distinct risk profiles not adequately captured by models developed in other geopolitical and industrial contexts. Future research should prioritize geographically diverse empirical studies to ensure generalizability.

4.3 Study Type and Methodological Profile

The methodological profile of the corpus (Fig. 4) is dominated by *Empirical Studies* ($n=10$, 52.6%), which encompass case studies, computational experiments with real or simulated datasets, and validation studies applying AI models

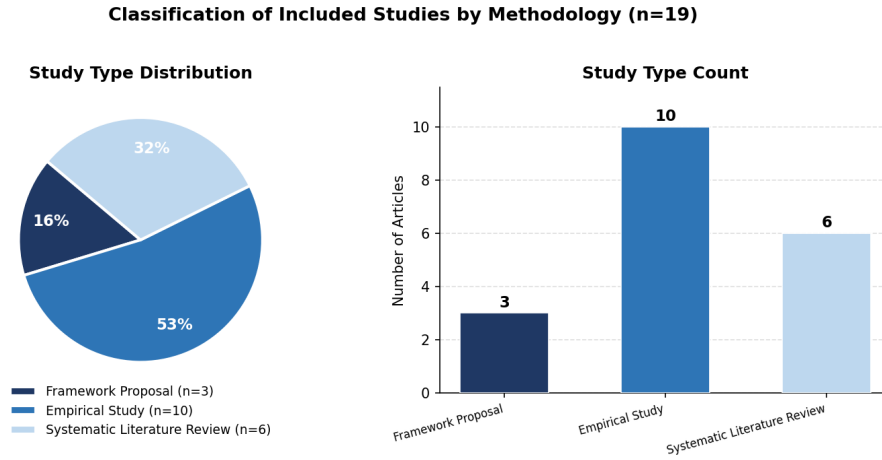


Fig. 4. Classification of included studies by methodology (n=19).

to supply chain data. This predominance signals that the field has progressed beyond purely theoretical discussion and is actively engaged in operationalizing AI solutions in applied settings.

Systematic Literature Reviews (SLR) constitute the second largest group (n=6, 31.6%), including bibliometric analyses and state-of-the-art reviews. The presence of six SLRs within a 19-article corpus is noteworthy, as it suggests that the community itself recognizes the need for synthesis and consolidation—a signal that the field is reaching a level of maturity where meta-analytical work is valued alongside primary research. This also implies that the knowledge base, while growing rapidly, remains fragmented, warranting ongoing integrative efforts such as the present review.

Framework Proposals (n=3, 15.8%) round out the corpus, representing conceptual contributions that map AI architectures and processes for SCRM without full empirical validation. These works serve as design blueprints for future implementations but highlight a gap: further empirical testing of proposed frameworks in real-world, longitudinal supply chain settings remains scarce.

4.4 Sectoral Coverage

The sectoral analysis (Fig. 5) exposes a significant limitation in the current literature: the majority of articles (n=11) adopt a *general* scope, applying AI-based SCRM frameworks without targeting a specific industry vertical. While such generality offers broad applicability, it simultaneously limits practical guidance for practitioners operating within sectors subject to idiosyncratic regulatory, operational, and risk environments.

Among sector-specific contributions, *E-commerce and Logistics* (n=5) represents the most investigated context, reflecting the real-time data availability and

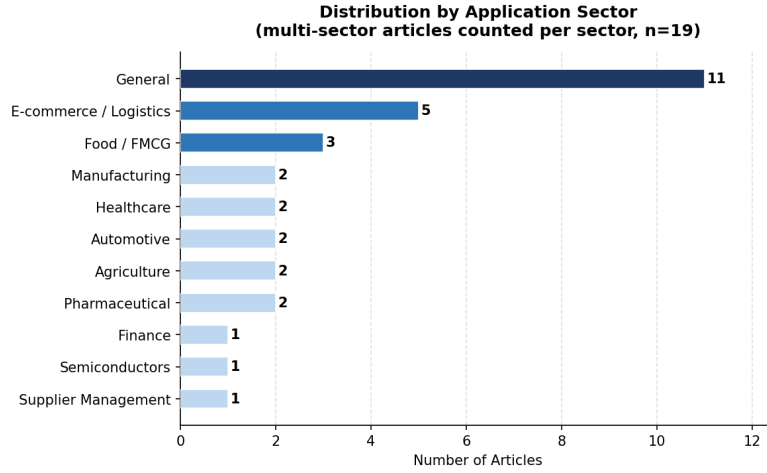


Fig. 5. Distribution of articles by application sector (multi-sector articles counted per sector, $n=19$).

transaction volume that make this sector particularly well-suited to AI-driven risk detection. *Food and FMCG*, *Healthcare*, *Automotive*, *Agriculture*, and *Pharmaceuticals* each appear in two articles, indicating nascent but emerging interest. Notably, sectors such as *Financial Services*, *Semiconductors*, and *Energy*—all of which face severe and well-documented supply chain vulnerabilities—are critically underrepresented.

This sectoral breadth gap is a substantive finding of this review. It implies that the AI techniques currently proposed may not be calibrated to sector-specific risk taxonomies, compliance constraints, or data infrastructures, and that domain-adapted AI models represent a priority avenue for future research.

4.5 Frequency of AI Techniques

The frequency analysis of AI techniques employed across the corpus (Fig. 6) reveals a clear hierarchy in adoption, with established, interpretable methods predominating over state-of-the-art generative or graph-based approaches.

Traditional Machine Learning (RF, SVM, GBM, and variants) is the most prevalent technique, appearing in seven articles. Its dominance is attributable to its strong performance on structured tabular data—the dominant format in ERP, WMS, and supplier management systems—as well as its relative computational efficiency and interpretability compared to deeper architectures. *NLP and Text Mining* ($n=6$) constitutes the second most frequent category, reflecting the growing recognition that unstructured data sources—social media, news feeds, regulatory announcements—contain early signals of supply chain stress that structured internal data cannot capture.

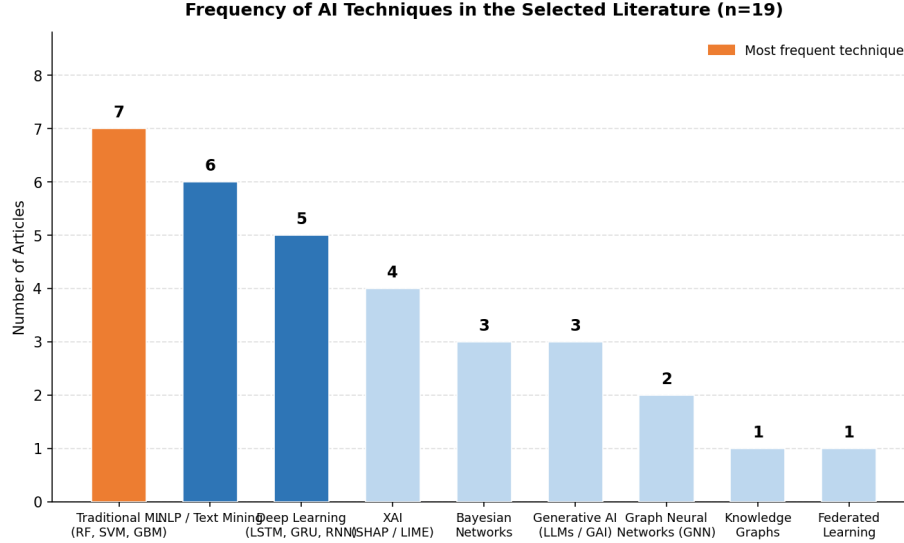


Fig. 6. Frequency of AI techniques identified across the selected literature (n=19).

Deep Learning architectures (LSTM, GRU, RNN; n=5) rank third, used primarily for sequential and time-series forecasting tasks. *Explainable AI* (SHAP, LIME; n=4) has gained meaningful traction, consistent with a growing demand for algorithmic transparency in high-stakes operational contexts. *Bayesian Networks* (n=3) and *Generative AI* (LLMs, GAI; n=3) appear with equal frequency, although their use cases differ markedly: the former for probabilistic multi-factor risk assessment, the latter for processing unstructured disruption signals.

At the lower end of adoption, *Graph Neural Networks* (n=2), *Knowledge Graphs* (n=1), and *Federated Learning* (n=1) remain niche despite their theoretical promise for modelling interconnected supply network topologies and privacy-preserving collaborative risk management, respectively. This underrepresentation is a pivotal finding: techniques most suited to the networked, relational nature of supply chains are the least deployed, suggesting that the field has not yet fully leveraged the structural characteristics of its own domain.

5 Content Analysis

5.1 Answer to RQ1 — How AI Mitigates Supply Chain Risks

The analysis of the 19 included articles reveals six distinct mechanisms through which artificial intelligence is applied to mitigate risks in supply chains. Early warning and forecasting constitutes the dominant approach, identified in eight articles (42%), followed by automation and decision support and resilience and recovery, each present in four articles (21%). Privacy-preserving collaboration,

explainability and transparency (XAI), and conceptual framework or review contributions each account for one article (5%) respectively.

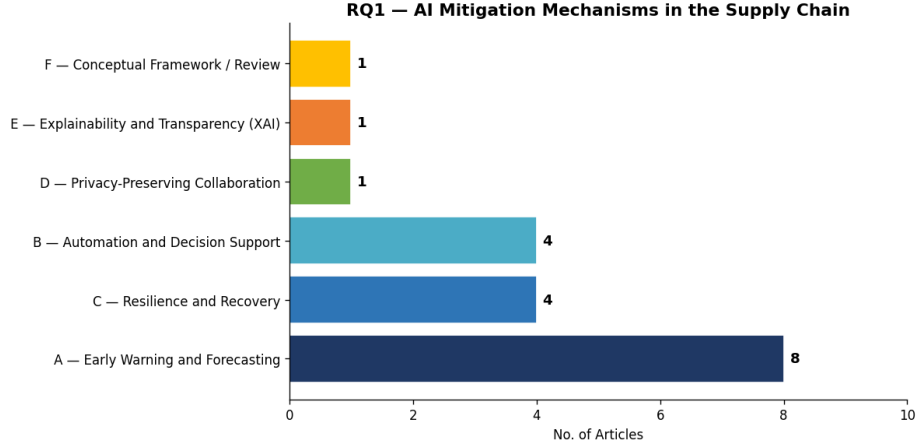


Fig. 7. Distribution of AI mitigation mechanisms across the 19 included articles (RQ1).

Early warning and forecasting is the most pervasive mitigation mechanism in the literature, reflecting a theoretical consensus that preemptive anticipation of supply chain disruptions is more effective than post-event recovery. Studies in this category deploy AI models trained on historical operational and environmental data to generate alerts before risks materialise. Gradient boosting architectures (including LightGBM, XGBoost, and Random Forest) are applied to predict delayed deliveries and payment anomalies [1], while deep learning frameworks combining GRU units, Bidirectional RNNs, and Transformer layers translate raw organisational data into probabilistic disruption forecasts [10]. Hybrid GNN-LSTM models capture both spatial supplier-relationship embeddings and temporal risk patterns simultaneously [5]. Machine learning and data mining underpin a collaborative early warning system for China’s manufacturing sector [17], and knowledge graph-based frameworks integrate Graph Neural Networks and weakly supervised learning for risk monitoring, early warning, and decision support [18]. Real-time text mining and sentiment analysis applied to social media streams extend the early warning paradigm to unstructured data sources, enabling detection of emerging disruptions such as port congestion and semiconductor shortages as they develop [4]. Review contributions document the broader transition from reactive to proactive risk management enabled by this class of AI application [3, ?].

Automation and decision support encompasses AI-driven systems that reduce human workload in risk-related decisions or automate responses to identified threats. Lin et al. [11] introduce the UNISONE framework, which combines a

Generative AI module for extracting disruption signals from unstructured text with Markov state modelling and Mixed Integer Programming for adaptive capacity reallocation. Explainable AI methods incorporating SHAP attribution and TOPSIS multi-criteria ranking automate and transparently support supplier selection decisions [8]. A multi-agent architecture integrating an LLM, an LSTM autoencoder, Random Forest, K-Means clustering, and Reinforcement Learning automates routing, anomaly detection, ESG scoring, and risk classification within a unified pipeline [16]. Clustering algorithms that segment suppliers into risk-performance profiles based on multiple operational indicators further exemplify AI-driven automation of vendor management decisions [7].

Resilience and recovery captures studies that use AI to model, simulate, or support the restoration of supply chain continuity following disruption. Fernando et al. [6] examine how machine learning informs manufacturing firms' adaptive responses to COVID-19 disruption, while Nayal et al. [12] investigate how AI adoption supports the agro-industry in sustaining operations under pandemic-induced supply chain stress. Nguyen et al. [14] demonstrate how Federated Machine Learning enables collaborative risk prediction across supply chain partners to support network-level resilience, and Naz et al. [13] systematically document AI as an enabler of supply chain resiliency in the post-COVID-19 context.

Privacy-preserving collaboration is addressed in one article. Nguyen et al. [14] show that Federated Machine Learning allows supply chain partners to collaboratively train risk prediction models (combining CNN and LSTM components) without sharing proprietary raw data, resolving a fundamental barrier to multi-stakeholder AI adoption.

Explainability and transparency (XAI) as a standalone mitigation mechanism is the focus of one systematic review. Nimmy et al. [15] comprehensively evaluate the suitability of XAI algorithms for supply chain operational risk management, establishing a methodological benchmark for transparent risk detection and positioning explainability as a first-class design requirement for applied AI in this domain.

Conceptual framework and review contributions provide the theoretical scaffolding on which applied studies build. Aldehayyat et al. [2] propose a digital transformation framework integrating AI and machine learning within an agile management paradigm, addressing the organisational prerequisites for AI-driven risk mitigation.

Overall, the dominance of early warning and forecasting reflects an underlying conviction in the literature that the most effective form of supply chain risk management is pre-emptive. The emergence of multi-agent and LLM-based architectures in 2025 articles suggests that the field is beginning to operationalise this logic at higher temporal resolution, moving from batch prediction toward continuous, automated risk surveillance.

5.2 Answer to RQ2 — Types of Supply Chain Risks Addressed

The cross-reference matrix produced from the 19 included articles reveals eight distinct risk type categories addressed through AI-based approaches: supplier

risk, disruption risk, logistics and transport risk, demand and inventory risk, cybersecurity risk, pandemic risk, geopolitical risk, and fraud and integrity risk. Coverage is unequal across these categories, with logistics and transport risk and disruption risk attracting the highest number of AI technique applications, while cybersecurity, geopolitical, and fraud risks remain markedly under-represented.

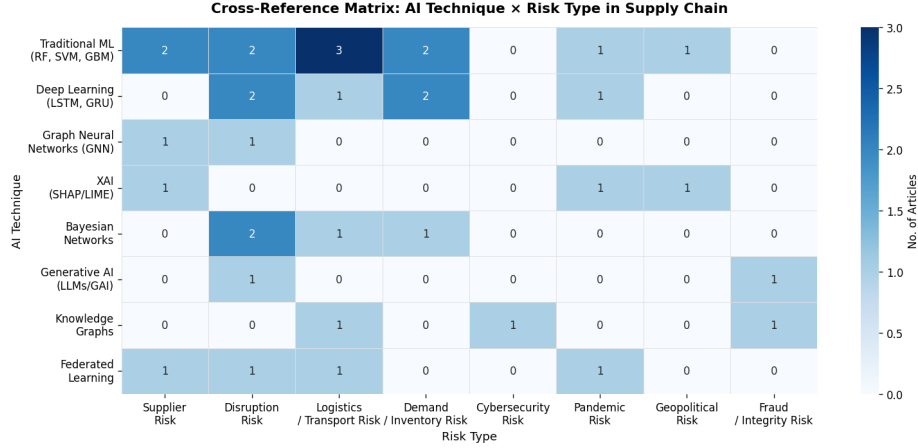


Fig. 8. Cross-reference matrix of AI techniques and supply chain risk types addressed in the included articles. Cell values indicate the number of articles addressing each combination.

Logistics and transport risk (encompassing transportation disruptions, port congestion, lead time variability, and distribution network failures) receives the broadest AI coverage across the corpus. Traditional machine learning methods (Random Forest, SVM, GBM) address this category in three articles, with additional coverage from Deep Learning, Bayesian Networks, Knowledge Graphs, and Federated Learning, reflecting the data richness and operational measurability of logistics processes [4, ?, ?].

Disruption risk is addressed by the widest range of AI technique categories (traditional ML, Deep Learning, Bayesian Networks, Generative AI, and Federated Learning), confirming it as the most cross-cutting risk type in the literature. This breadth reflects the scale of academic response to the COVID-19 pandemic and subsequent geopolitical instability, which exposed the fragility of global supply networks to macro-level shocks [10, ?, ?, ?].

Supplier risk (covering supplier failure, quality issues, delivery delays, and credit default) is addressed predominantly through traditional ML methods and GNNs, which are particularly suited to modelling network-level interdependencies among suppliers [5, ?, ?].

Demand and inventory risk, arising from demand volatility, forecast errors, and inventory misalignment, is addressed by traditional ML and Deep Learning

in two articles each, reflecting the long-standing application of statistical learning to demand forecasting problems [1, ?].

Pandemic risk receives dedicated attention in three articles through traditional ML, XAI, and Federated Learning, confirming the COVID-19 pandemic as a structuring event for this research stream [12, ?, ?].

Geopolitical risk is addressed in two articles through traditional ML and XAI-based approaches [2, ?]. *Fraud and integrity risk* appears in two articles deploying Generative AI and Knowledge Graphs [11, ?]. *Cybersecurity risk* is the least studied category, identified in only one article through a Knowledge Graph-based approach [18], representing a critical gap given the increasing digitalisation of supply chain operations.

5.3 Answer to RQ3 — AI Techniques Used

The methodological landscape of the 19 included articles encompasses eight distinct AI technique categories. Traditional machine learning (comprising Random Forest, Support Vector Machines, and Gradient Boosting Methods) is the most widely deployed category, applied across the broadest range of risk types. Deep Learning (LSTM, GRU), Graph Neural Networks (GNN), XAI (SHAP/LIME), Bayesian Networks, Generative AI (LLMs), Knowledge Graphs, and Federated Learning constitute the remaining categories, reflecting a methodologically diverse corpus.

Traditional machine learning (Random Forest, SVM, GBM) addresses supplier risk (2 articles), disruption risk (2 articles), logistics and transport risk (3 articles), demand and inventory risk (2 articles), pandemic risk (1 article), and geopolitical risk (1 article), demonstrating the broadest risk coverage of any single technique category [1, ?, ?, ?]. The versatility of these methods for structured, tabular supply chain data explains their continued prevalence despite the emergence of more complex architectures.

Deep Learning (LSTM, GRU) is applied to disruption risk (2 articles), logistics and transport risk (1 article), demand and inventory risk (2 articles), and pandemic risk (1 article) [10, ?]. The temporal modelling capabilities of recurrent architectures make them particularly suited to supply chain time-series data, where historical sequences encode the dynamics of disruption propagation.

Graph Neural Networks address supplier risk (1 article) and logistics and transport risk (1 article) [18, ?], reflecting the natural fit between graph-structured models and the network topology of supply chains, where risks propagate through supplier-buyer relationships.

XAI (SHAP/LIME) is applied to supplier risk (1 article), pandemic risk (1 article), and geopolitical risk (1 article) [8, ?], confirming the growing importance of interpretability as a design requirement for AI systems deployed in high-stakes supply chain decisions.

Bayesian Networks address disruption risk (2 articles), logistics and transport risk (1 article), and demand and inventory risk (1 article) [3, ?], providing probabilistic modelling capabilities well-suited to uncertainty quantification under incomplete information.

Generative AI (LLMs) appears in two articles addressing disruption risk and fraud and integrity risk [11, ?], representing the most recent technical development in the corpus and signalling the entry of large language model capabilities into supply chain risk management.

Knowledge Graphs address logistics and transport risk (1 article), cybersecurity risk (1 article), and fraud and integrity risk (1 article) [18], offering structured knowledge representation capabilities for complex, multi-entity risk environments.

Federated Learning addresses supplier risk (1 article), disruption risk (1 article), logistics and transport risk (1 article), and pandemic risk (1 article) [14], enabling privacy-preserving collaborative risk modelling across organisational boundaries.

A temporal trend is discernible: articles published in 2022 and 2023 predominantly rely on traditional ML and review-based methods, while 2024 and 2025 contributions increasingly deploy hybrid architectures combining Deep Learning, GNNs, Generative AI, and formal optimisation. The appearance of LLMs in two 2025 articles suggests that generative AI capabilities are beginning to be operationalised for supply chain risk management, a trend likely to intensify as domain-specific model fine-tuning becomes more accessible.

5.4 Complementary Analysis

Sector concentration. The majority of included articles propose general or sector-agnostic supply chain models, limiting the ability to draw sector-specific conclusions. Where specific industries are targeted, manufacturing receives the most attention [17, ?], followed by logistics and transportation [9, ?], agriculture [12], e-commerce and contract logistics [1], and multi-sector contexts including pharmaceuticals, food, and semiconductors [11]. Healthcare, energy, and public sector supply chains are absent from the corpus, representing sectors where AI-based risk management frameworks remain underdeveloped.

Geographic distribution. Indian institutions are affiliated with six articles [3, ?, ?, ?, ?, ?], constituting the most active national research community in the corpus. China contributes two articles [18, ?] and Vietnam two articles [9, ?], with the remaining studies originating from multi-institutional collaborations across Taiwan, Japan, Jordan, the United Kingdom, France, and Brazil. Primary contributions from North America and continental Europe are absent, despite these regions' substantial supply chain research traditions.

Temporal evolution. Publication volume increases steadily from two articles in 2022 to four in 2025, with one article already slated for 2026, confirming a clear upward trajectory. The COVID-19 pandemic serves as the principal inflection point, with the most technically sophisticated contributions (deploying LLMs, multi-agent systems, and advanced hybrid architectures) concentrated in 2025.

6 Discussion

6.1 Synthesis of Key Findings

Collectively, the three research questions reveal a field in active and accelerating development, characterised by increasing methodological sophistication and a clear orientation toward preemptive risk management over reactive recovery. The literature converges strongly on early warning and forecasting as the primary mode of AI-enabled risk mitigation [3, ?, ?], reflecting a theoretical consensus that the most effective supply chain risk management intervenes before disruptions propagate through the network.

The cross-reference matrix (Fig. 8) reveals structural patterns in the relationship between AI techniques and risk types. Traditional machine learning dominates across the broadest risk portfolio, reflecting its versatility for structured operational data. Deep Learning and hybrid architectures concentrate on disruption, logistics, and demand risks, which are characterised by temporal complexity and high data volume. Generative AI and Knowledge Graphs are emerging in fraud and integrity risk contexts, where the capacity to reason over unstructured information and relational knowledge is particularly valuable. XAI techniques are distributed across supplier, pandemic, and geopolitical risks, confirming that high-stakes, low-frequency risk decisions create the strongest demand for interpretable AI outputs.

The dominance of logistics and transport risk and disruption risk in the cross-reference matrix, relative to the near-absence of cybersecurity, geopolitical, and fraud risks, reveals a field that is strongly responsive to operationally visible and historically recent disruptions, while exhibiting limited engagement with risks that are either diffuse, novel, or difficult to quantify from structured data alone.

6.2 Emerging Trends

Transition from single-model to multi-agent architectures. The most architecturally ambitious contributions of 2025 deploy systems in which multiple specialised AI components perform distinct functions within a unified pipeline. Rajesh et al. [16] integrate an LLM, Random Forest, K-Means, an LSTM autoencoder, and Reinforcement Learning within a single multi-agent framework. Lin et al. [11] combine Generative AI, Markov modelling, and Mixed Integer Programming. This represents a qualitative departure from the single-model approaches predominant in 2022 and 2023 articles and suggests a trajectory toward integrated supply chain risk intelligence platforms.

Entry of Generative AI and LLMs into supply chain risk management. Two 2025 articles deploy Large Language Models as active components of risk management architectures [11, ?], a development with no precedent in earlier articles of the corpus. The capability of LLMs to extract structured risk signals from unstructured text (news, regulatory updates, supplier communications) addresses a longstanding limitation of traditional ML approaches, which require structured input data. This trend is likely to intensify as domain-specific fine-tuning and

retrieval-augmented generation techniques lower the cost of deploying LLMs in operational contexts.

Growth of explainability as a design requirement. The increasing prevalence of SHAP and LIME-based interpretability frameworks (evidenced in articles by Adom et al. [1], Kumar and Kumar [8], Li et al. [10], and the systematic review by Nimmy et al. [15]) signals that explainability has transitioned from an optional enhancement to a first-class design requirement. This reflects the practical reality that supply chain risk decisions carry significant financial and operational consequences, requiring not only accurate predictions but transparent justifications that managers can act upon and audit.

Post-COVID-19 acceleration and diversification of risk categories studied. The temporal distribution confirms the COVID-19 pandemic as a research accelerant, generating both the empirical urgency and the data availability needed to develop and validate AI-based risk management systems. Post-2022 articles begin to diversify beyond pandemic risk into geopolitical risk [2], fraud and integrity risk [11], and ESG-related risk [16], suggesting that the field is broadening its risk aperture beyond the immediate disruption context.

6.3 Identified Gaps

The research gaps identified across the corpus are summarised in Fig. 9 and discussed below.

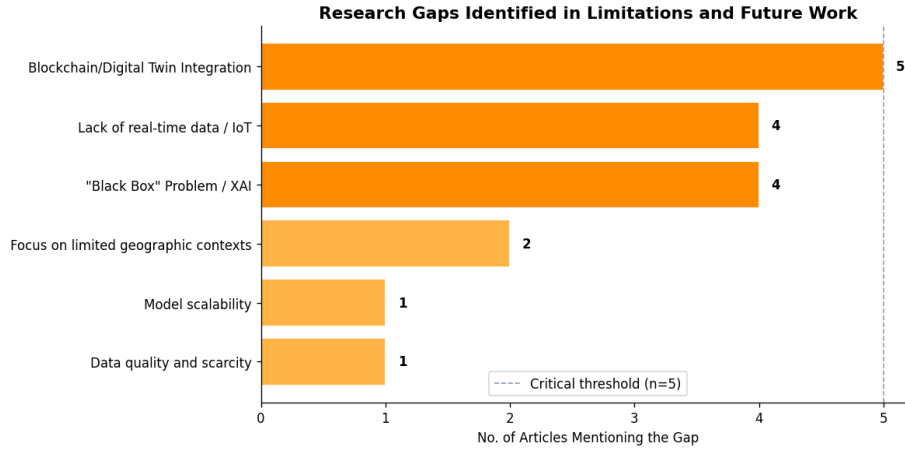


Fig. 9. Research gaps identified across the included articles, ranked by frequency of mention.

Blockchain and Digital Twin integration is the most frequently identified gap, mentioned in five articles. The literature consistently notes that AI-based risk management systems would benefit substantially from integration with blockchain

for immutable audit trails and Digital Twin environments for real-time simulation of disruption scenarios [6, ?, ?]. The absence of empirical studies implementing such integrations at operational scale represents the most prominent frontier for future research.

Lack of real-time data and IoT integration is cited in four articles as a barrier to the operationalisation of AI risk management frameworks. Many proposed models are validated on historical batch datasets rather than live operational streams, limiting their applicability for real-time early warning [4, ?]. Future research should prioritise architectures that ingest IoT sensor data, RFID streams, and other real-time supply chain telemetry.

The black box problem and the need for XAI is identified in four articles as a critical barrier to managerial adoption [15, ?, ?, ?]. Despite the growing prevalence of SHAP-based interpretability in recent contributions, the majority of proposed AI systems still lack sufficient transparency for deployment in regulated or safety-critical supply chain environments. Developing explainability frameworks specifically tailored to supply chain risk contexts remains an open research challenge.

Focus on limited geographic contexts is noted in two articles [17, ?]. The concentration of empirical studies in Asian manufacturing and logistics contexts limits the generalisability of findings to supply chains operating under different regulatory frameworks, infrastructure conditions, and risk exposure profiles. Studies conducted in European, North American, and developing-economy contexts are needed to establish the cross-contextual validity of AI-based risk management frameworks.

Model scalability and *data quality and scarcity* are each identified in one article as constraints on the practical deployment of AI systems [10, ?]. Supply chain disruption events are rare and high-impact, creating class-imbalance problems that undermine the reliability of models trained on historical data. Techniques for few-shot learning, synthetic data generation, and transfer learning across supply chain domains represent promising avenues for addressing these constraints.

6.4 Limitations of This Review

This review is subject to several limitations that should be considered when interpreting its findings. First, the literature search was restricted to two databases (IEEE Xplore and Web of Science), which, while chosen for their complementary technical and multidisciplinary coverage, may have excluded relevant contributions indexed in Scopus, the ACM Digital Library, or domain-specific repositories. Second, the time restriction to 2019 and 2025 excludes foundational work published prior to 2019 that may underpin more recent methodological contributions. Third, the language restriction to English and Portuguese may introduce a systematic bias against research communities publishing in Mandarin, Arabic, or other languages, which is particularly relevant given the geographic concentration of the corpus in Asia. Finally, the classification of AI techniques and risk types relied in some instances on abstract-level metadata rather than deep

full-text algorithmic audits, which may have introduced minor categorisation imprecision.

7 Conclusion

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